LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations

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SECTION A

Q1. Which of the following is a correctly balanced chemical equation? **[1]**

Answer: (b) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$. Both sides have 3 Fe, 8 H and 4 O.

Q2. When dilute hydrochloric acid is added to zinc granules, the gas evolved is: **[1]**

Answer: (b) hydrogen. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Q3. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ takes place in sunlight. This is an example of: **[1]**

Answer: (c) photolytic decomposition (decomposition by light)

Q4. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, the substance that gets oxidised is: **[1]**

Answer: (b) H_2. It gains oxygen to form H_2O.

Q5. Which of the following is an exothermic process? **[1]**

Answer: (b) Respiration. Glucose is oxidised and energy is released.

Q6. **Assertion (A):** The blue colour of copper sulphate solution fades when an iron nail is dipped in it. **Reason (R):** Iron is more reactive than copper and… **[1]**

Answer: (a). $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$. The solution turns from blue to light green.

SECTION B

Q7. Why should a magnesium ribbon be cleaned with sandpaper before burning it in air? Write the balanced equation for its burning. **[2]**

Answer: The ribbon has a layer of magnesium oxide (formed by reaction with air) on its surface, which stops it from burning easily. Sandpaper removes this layer. $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

- Reason (oxide layer removed): 1
- Balanced equation: 1

Q8. What is a precipitation reaction? Give one example with a balanced chemical equation. **[2]**

Answer: A reaction in which an insoluble solid (precipitate) is formed. $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$, which gives a white precipitate of $BaSO_4$.

- Definition: 1
- Correct balanced example: 1

OR, answer: Nitrogen is unreactive and replaces oxygen in the packet, so the fats and oils in the chips are not oxidised. This prevents rancidity (change in smell and taste).

- Nitrogen prevents oxidation of fats/oils: 1
- Names rancidity: 1

SECTION C

Q9. A solution of a substance 'X' is used for whitewashing walls. **[3]**

(i) Calcium oxide (quicklime), CaO [1]

(ii) $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + heat$ [1]

(iii) $Ca(OH)_2$ reacts slowly with CO_2 in air to form a thin, shiny layer of calcium carbonate: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ [1]

Q10. State the type of each of the following reactions: **[3]**

(i) Displacement reaction [1]

(ii) Decomposition reaction (electrolytic decomposition) [1]

(iii) Double displacement (precipitation) reaction, which gives a yellow precipitate of PbI_2 [1]

SECTION D

Q11. Answer the following: **[5]**

(i) A reaction in which one substance is oxidised and another is reduced at the same time. [1]

(ii) C is oxidised (it gains oxygen to form CO). ZnO is reduced (it loses oxygen to form Zn). [2]

- Each correct identification with reason: 1

(iii) Corrosion is the slow eating away of a metal when it is attacked by substances around it such as moisture, oxygen and acids (e.g. rusting of iron). Prevention (any two): painting, oiling/greasing, galvanising, chrome plating, making alloys. [2]

- Definition: 1
- Any two methods: ½ + ½

OR, answer: (a) The green crystals first lose water and then turn reddish-brown, and a smell of burning sulphur is noticed. $2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$, on heating. This is thermal decomposition. (b) In respiration glucose combines with oxygen in our cells and energy is released: $C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$ + energy.

- (a) observation 1, equation 1, type 1
- (b) reason 1, equation 1

SECTION E

Q12. In a class activity, students heated lead nitrate powder in a boiling tube. Brown fumes were given off and a yellowish residue was left behind. In… **[4]**

(i) Nitrogen dioxide (NO_2). $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ [1]

(ii) Thermal decomposition reaction [1]

(iii) Basic copper carbonate. Copper reacts slowly with moist carbon dioxide (and oxygen) in the air. This is corrosion of copper. [2]

- Names basic copper carbonate: 1
- Reason (reaction with moist CO_2/air, corrosion): 1